Convex Optimization

CS5491: Artificial Intelligence
ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course and Prof. Boddeti's AI Course

TODAY

Recognize convex sets, functions, and optimization problems.

Distinguish existence, global optimality, and uniqueness.

Calculate gradient-descent updates and explain the role of step size.

LP, IP, AND CONVEX OPTIMIZATION

LP: a linear objective over a convex feasible region.

Adding integer restrictions generally makes the feasible set nonconvex.

A convex objective need not be linear. Its minimum can lie inside the feasible region.

Example: minimize t^2 on $[-1,1]$. The optimum is $t=0$, not a vertex.

CONVEX OPTIMIZATION: DEFINITION

$$\min_{\mathbf{x}\in\mathcal{F}} f(\mathbf{x})$$

The feasible set $\mathcal{F}$ is convex.

The objective f is convex on $\mathcal{F}$.

Both conditions matter. We will examine each one.

CONVEX COMBINATION

A point on the line segment joining $\mathbf{x}$ and $\mathbf{y}$:
$$\mathbf{z}=\theta\mathbf{x}+(1-\theta)\mathbf{y},\qquad 0\le\theta\le1.$$

$\theta=0$ or 1 gives an endpoint; $\theta=1/2$ gives the midpoint.

For $\mathbf{x}=(0,0)$ and $\mathbf{y}=(2,4)$, their midpoint is $(1,2)$.

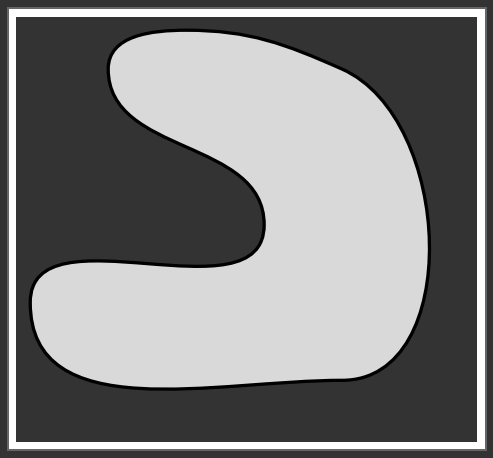
When $0<\theta<1$, this is a strict convex combination.

CONVEX SETS

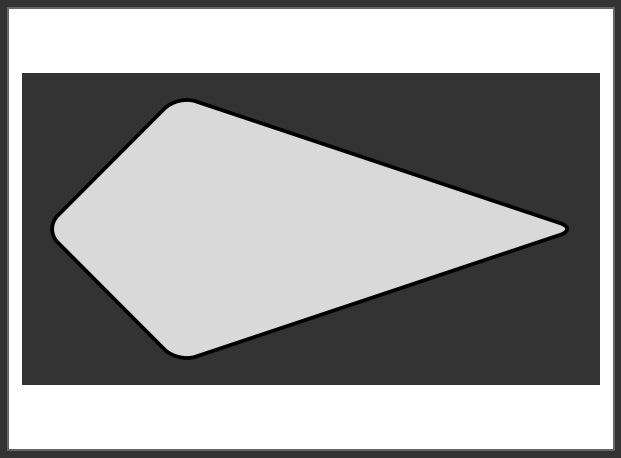
Every line segment joining two points in the set stays inside the set.

$$\mathbf{x},\mathbf{y}\in\mathcal{F},\ 0\le\theta\le1 \ \Longrightarrow\ \theta\mathbf{x}+(1-\theta)\mathbf{y}\in\mathcal{F}.$$

Nonconvex set



Convex set



QUIZ: CONVEX SETS

For $n\ge1$ and convex $\mathcal{F}_1,\mathcal{F}_2$, which sets must be convex?

Set	Answer and reason
$\mathbb{R}^n$	Convex: every segment stays in $\mathbb{R}^n$.
$\emptyset$	Convex: no pair violates the definition.
$\{\mathbf{x}_0\}$	Convex: every combination equals $\mathbf{x}_0$.
$\mathcal{F}_1\cap\mathcal{F}_2$	Convex if both sets are convex.
$\mathcal{F}_1\cup\mathcal{F}_2$	Not necessarily: $\{0\}\cup\{2\}$ omits 1.

Set	Answer and reason
$\mathbb{Z}^n$	Not convex: a midpoint need not be integral.

CONVEX FUNCTIONS

The value at a convex combination is no greater than the corresponding weighted average of values.

For a convex domain $\mathcal{F}$, all $\mathbf{x},\mathbf{y}\in\mathcal{F}$ and $\theta\in[0,1]$:

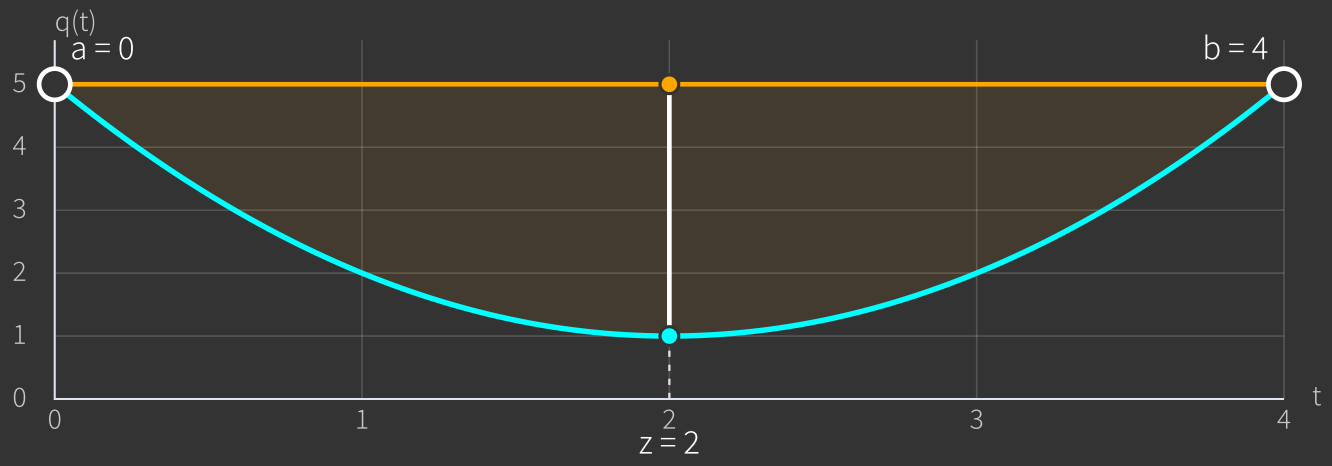
$$\color{cyan}{f(\theta\mathbf{x}+(1-\theta)\mathbf{y})} \le \color{orange}{\theta f(\mathbf{x})+(1-\theta)f(\mathbf{y})}.$$

Equality is allowed. In particular, every affine function is convex.

Strict convexity requires $<$ whenever $\mathbf{x}\ne\mathbf{y}$ and $0<\theta<1$.

THE GRAPH LIES BELOW ITS CHORD

$q(t)=(t-2)^2+1$. Drag the endpoints and vary θ.



Endpoint a: 0 Endpoint b: 4 Weight θ: 0.5 Reset

$z=\theta a+(1-\theta)b=$ 2

$q(z)=$ 1 $\le$ $\theta q(a)+(1-\theta)q(b)=$ 5

For every $a,b\in\mathbb{R}$ and $\theta\in[0,1]$:

$\underbrace{\theta q(a)+(1-\theta)q(b)-q(z)}_{\text{gap}}=\theta(1-\theta)(a-b)^2\ge0.$

Current gap: 4 • Strict inequality

CHECKING CONVEXITY

Apply the defining inequality.

Use a nonnegative weighted sum of convex functions: $$f(\mathbf{x})=\sum_i w_i f_i(\mathbf{x}),\qquad w_i\ge0.$$

Compose a convex function with an affine map: $$f(\mathbf{x})=g(A\mathbf{x}+\mathbf{b}).$$

THE SECOND-DERIVATIVE TEST

For a twice continuously differentiable function on an open interval I:

$$f\text{ is convex on }I\quad\Longleftrightarrow\quad f''(t)\ge0\text{ for all }t\in I.$$

Example: $q(t)=(t-2)^2+1$, for all real t.

$$q'(t)=2(t-2),\qquad q''(t)=2>0.$$

Therefore q is convex (in fact, strictly convex).

THE GRADIENT

For a differentiable $f:\mathbb{R}^n\to\mathbb{R}$:
$$\nabla f(\mathbf{x})= \begin{bmatrix}\partial f/\partial x_1\\\vdots\\\partial f/\partial x_n\end{bmatrix} \in\mathbb{R}^n.$$

A vector of partial derivatives, evaluated at $\mathbf{x}$.

When nonzero, it points in the direction of steepest local increase under the Euclidean norm.

When the gradient is nonzero, its negative is a local descent direction.

THE HESSIAN TEST

Let $U\subseteq\mathbb{R}^n$ be open and convex, and let $f\in C^2(U)$.

$$H(\mathbf{x})=\nabla^2 f(\mathbf{x}),\qquad H_{ij}(\mathbf{x})=\frac{\partial^2 f}{\partial x_i\,\partial x_j}.$$

$$f\text{ is convex on }U \quad\Longleftrightarrow\quad H(\mathbf{x})\succeq 0\text{ for every }\mathbf{x}\in U.$$

Positive semidefinite means $\mathbf{z}^T H(\mathbf{x})\mathbf{z}\ge 0$ for every $\mathbf{z}$.

At each $\mathbf{x}$, this is equivalent to all eigenvalues of the symmetric matrix $H(\mathbf{x})$ being nonnegative.

A POSITIVE-SEMIDEFINITE HESSIAN

$$F(u,v)=(u+v-1)^2.$$
$$\nabla F=2(u+v-1)\begin{bmatrix}1\\1\end{bmatrix},\qquad H=\begin{bmatrix}2&2\\2&2\end{bmatrix}.$$
$$\mathbf{z}^T H\mathbf{z}=2(z_1+z_2)^2\ge0.$$
Thus F is convex on $\mathbb{R}^2$. The Hessian has eigenvalues 0 and 4.

CONCAVITY AND CONVEXITY

f is concave exactly when $-f$ is convex.
$$f(\theta\mathbf{x}+(1-\theta)\mathbf{y}) \ge\theta f(\mathbf{x})+(1-\theta)f(\mathbf{y}).$$
Maximizing a concave function over a convex set is equivalent to a convex minimization problem:
$$\max_{\mathbf{x}\in\mathcal{F}}f(\mathbf{x}) \quad\Longleftrightarrow\quad\min_{\mathbf{x}\in\mathca [-f(\mathbf{x})].$$

QUIZ: CONVEX FUNCTIONS

Function and domain	Answer and reason
e^{at}, $t\in\mathbb{R}$, $a\in\mathbb{R}$	Convex: $f''=a^2e^{at}\ge0$.
$\log t$, $t>0$	Concave, not convex: $f''=-1/t^2$.
$\|\mathbf{x}\|_2$, $\mathbf{x}\in\mathbb{R}^n$	Convex; not differentiable at $\mathbf{0}$.
$\mathbf{x}^T A\mathbf{y}$, jointly in $(\mathbf{x},\mathbf{y})$	Convex iff $A=0$; otherwise neither convex nor concave.
t^3, $t\in\mathbb{R}$	Neither: $f''(t)=6t$ changes sign.

LOCAL AND GLOBAL MINIMA

A global minimum occurs at a feasible point whose objective value is no greater than at any feasible point.

A local minimum occurs at a feasible point whose objective value is no greater than at nearby feasible points.

For a convex optimization problem, every local minimum is global.
If a better feasible point existed, moving a little toward it would already improve the objective.

EXISTENCE AND UNIQUENESS

Convexity alone guarantees neither a minimum nor a unique point attaining it.

Convex problem	What happens?
$\min_{t\in\mathbb{R}}[(t-2)^2+1]$	Minimum attained only at $t=2$.
$\min_{t\in\mathbb{R}}e^t$	Infimum 0, but no attained minimum.
$\min_{0\le u,v\le1}(u+v-1)^2$	Minimum attained at infinitely many points.

With strict convexity, *at most one point* attains the minimum; existence is a separate question.

FINDING ALL MINIMA

$$\min_{0\le u,v\le1} F(u,v),\qquad F(u,v)=(u+v-1)^2.$$

A square is nonnegative: $F(u,v)\ge0$.

Equality holds exactly when $u+v=1$.

$$F^*=0,\qquad \operatorname{argmin}F=\{(u,1-u):0\le u\le1\}.$$
Minima occur at $(0,1)$, $(1,0)$, and every point on the segment between them.

GRADIENT DESCENT

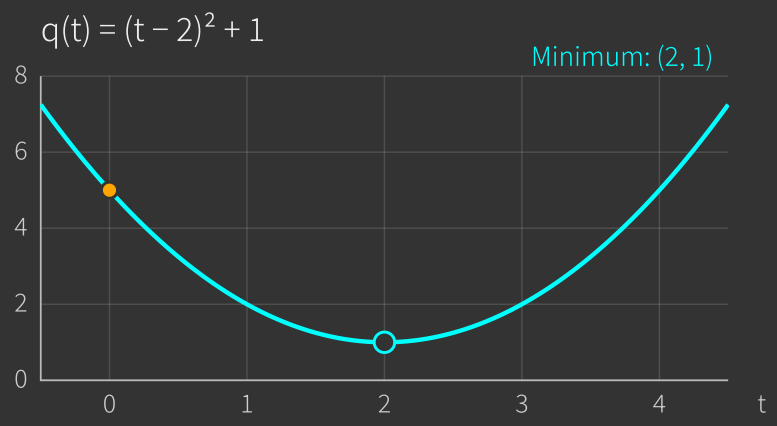
For an unconstrained, differentiable objective:
$$\mathbf{x}_{k+1}=\mathbf{x}_k-\alpha_k\nabla f(\mathbf{x}_k),\qquad\alpha_k>0.$$

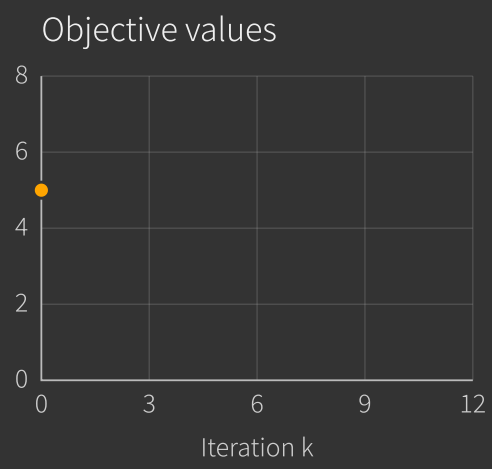
1. Choose an initial point $\mathbf{x}_0$.
2. Evaluate the gradient at the current point.
3. Take a step in the negative-gradient direction.
4. Repeat until a stopping criterion is satisfied.

A GRADIENT-DESCENT UPDATE

$$q(t)=(t-2)^2+1,\qquad t_0=0,\qquad\alpha=\tfrac14.$$
$$q'(t)=2(t-2),\qquad q'(0)=-4.$$
$$t_1=0-\tfrac14(-4)=1,\qquad q(t_1)=2<q(t_0)=5.$$
$$t_2=1-\tfrac14(-2)=\tfrac32,\qquad q(t_2)=\tfrac54.$$
The global minimum occurs at $t^*=2$, with value $q(t^*)=1$.

ITERATES AND OBJECTIVE VALUES





Take a step to see the parameter update.

Start t_0: 0 Step size α: 0.25 Back Next step Restart

$k=$ 0 $t_k=$ 0 $q(t_k)=$ 5 $q'(t_k)=$ −4

Next: $t_{k+1}=t_k-2\alpha(t_k-2)=$ 1

Approaches the minimum from one side.

Orange: current iterate. Values rounded.

STEP SIZE MATTERS

For $q(t)=(t-2)^2+1$, the minimum occurs at $t^*=2$.

$$\begin{aligned} t_{k+1}&=t_k-2\alpha(t_k-2) &&\text{gradient step}\\ t_{k+1}-2&=(t_k-2)-2\alpha(t_k-2) &&\text{subtract }2\\ \color{cyan}{t_{k+1}-2}&=\color{orange}{(1-2\alpha)}\color{cyan}{(t_k-2)} &&\text{factor} \end{aligned}$$

Each step multiplies the signed error t_k-2 by $1-2\alpha$.

α	Factor $1-2\alpha$	Starting at $t_0=0$	Behavior
$1/4$	$1/2$	$0,1,3/2,\ldots$	Converges to 2.
1	-1	$0,4,0,4,\ldots$	Oscillates.
$5/4$	$-3/2$	$0,5,-5/2,\ldots$	Diverges.

Distance from 2 shrinks when $|1-2\alpha|<1$, that is, $0<\alpha<1$.

STOPPING AND OPTIMALITY

For unconstrained differentiable convex f, $\nabla f(\mathbf{x})=0$ certifies a global minimum.

In computation, monitor the gradient norm, objective changes, and an iteration limit.

A small update alone may reflect a tiny step size.

For $f(t)=t^2$, $t_0=100$, and $\alpha=10^{-12}$:
$$|t_1-t_0|=2\times10^{-10},\qquad f(t_1)\approx10000.$$

PRACTICE: ONE STEP AND THE OPTIMUM

$$r(t)=(t+1)^2+3,\qquad t_0=1,\qquad\alpha=\tfrac14.$$

Calculate	Answer
$r'(t)$ and $r''(t)$	$2(t+1)$ and 2; hence convex.
$r'(t_0)$	$r'(1)=4$.
The next iterate t_1	$1-\tfrac14(4)=0$.
Global minimum: location and value	$t^*=-1$, $r(t^*)=3$.

PROJECTED GRADIENT DESCENT

Optional background; not assessed in the midterm.

For a nonempty, closed, convex feasible set $\mathcal{F}$:

$$P_{\mathcal{F}}(\mathbf{z})=\operatorname{argmin}_{\mathbf{u}\in\mathcal{F}}\|\mathbf{z}-\mathbf{u}\|_2^2.$$

$$\mathbf{x}_{k+1}=P_{\mathcal{F}}\bigl(\mathbf{x}_k-\alpha_k\nabla f(\mathbf{x}_k)\bigr).$$

Start with $\mathbf{x}_0\in\mathcal{F}$. Take a gradient step, then project back to the feasible set.

SOLUTION METHODS AND SCOPE

Core: recognize convexity and interpret global minima.

Core: solve $\nabla f=0$ and calculate gradient-descent updates for simple unconstrained examples.

Optional background: projected gradient descent.

Newton, interior-point, and subgradient methods are outside this lecture's assessed techniques.

APPLYING CONVEX OPTIMIZATION

1. Define the variables, objective, and feasible set.
2. Establish convexity of both objective and feasible set.
3. Solve the model and check feasibility and solver status.
4. Interpret the solution in the original problem.

Tools include CVXPY, CVX, and CVXOPT. Modeling systems such as CVXPY recognize specific convex forms; a mathematically convex expression may need rewriting.

KEY POINTS

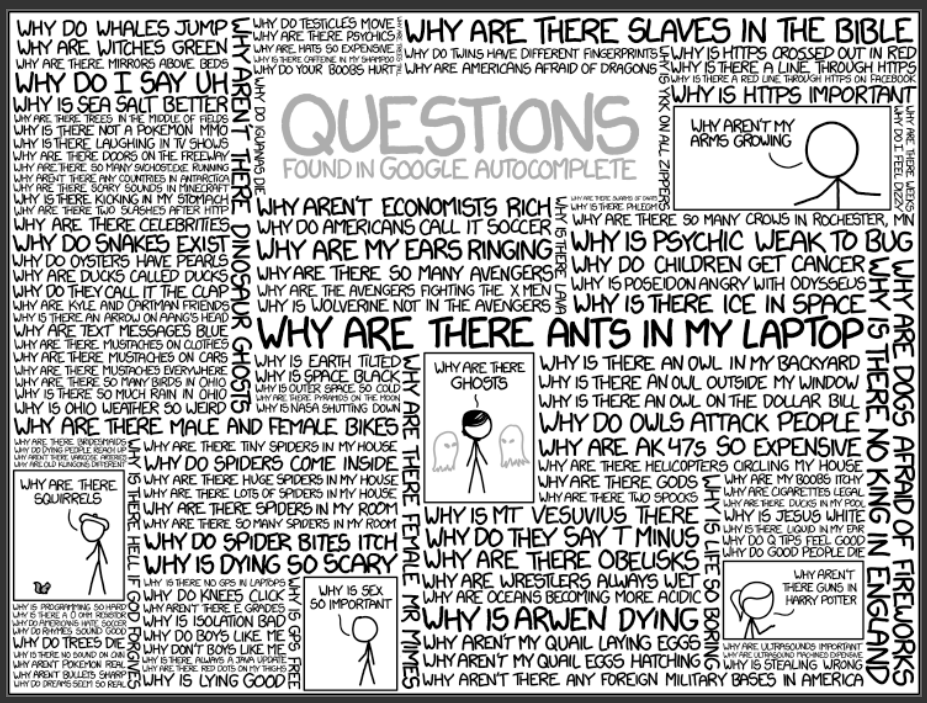
Convex sets contain their line segments; convex functions lie below their chords.

Local minima are global, but existence and uniqueness are separate issues.

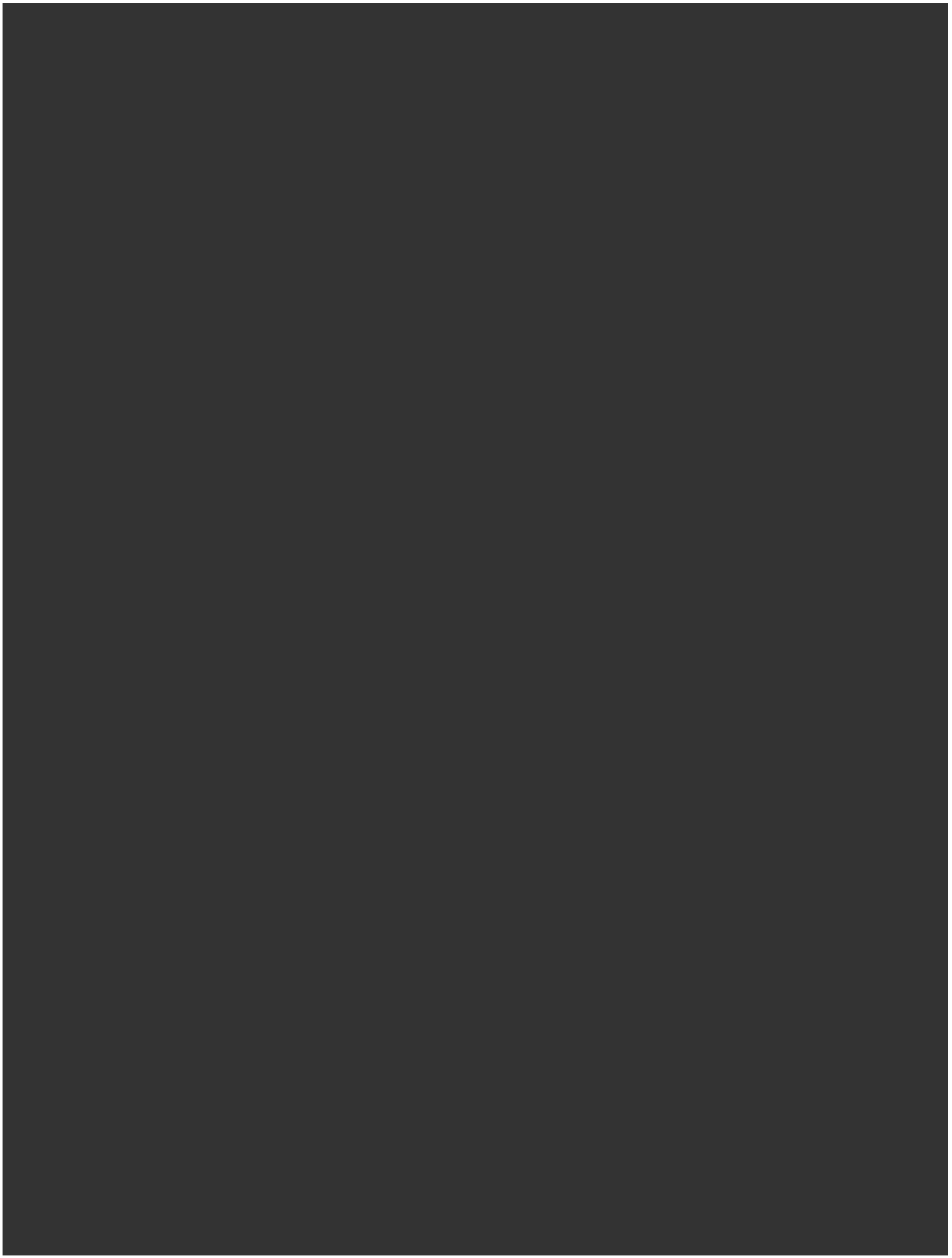
Gradient descent updates parameters using the negative gradient; step size affects convergence.

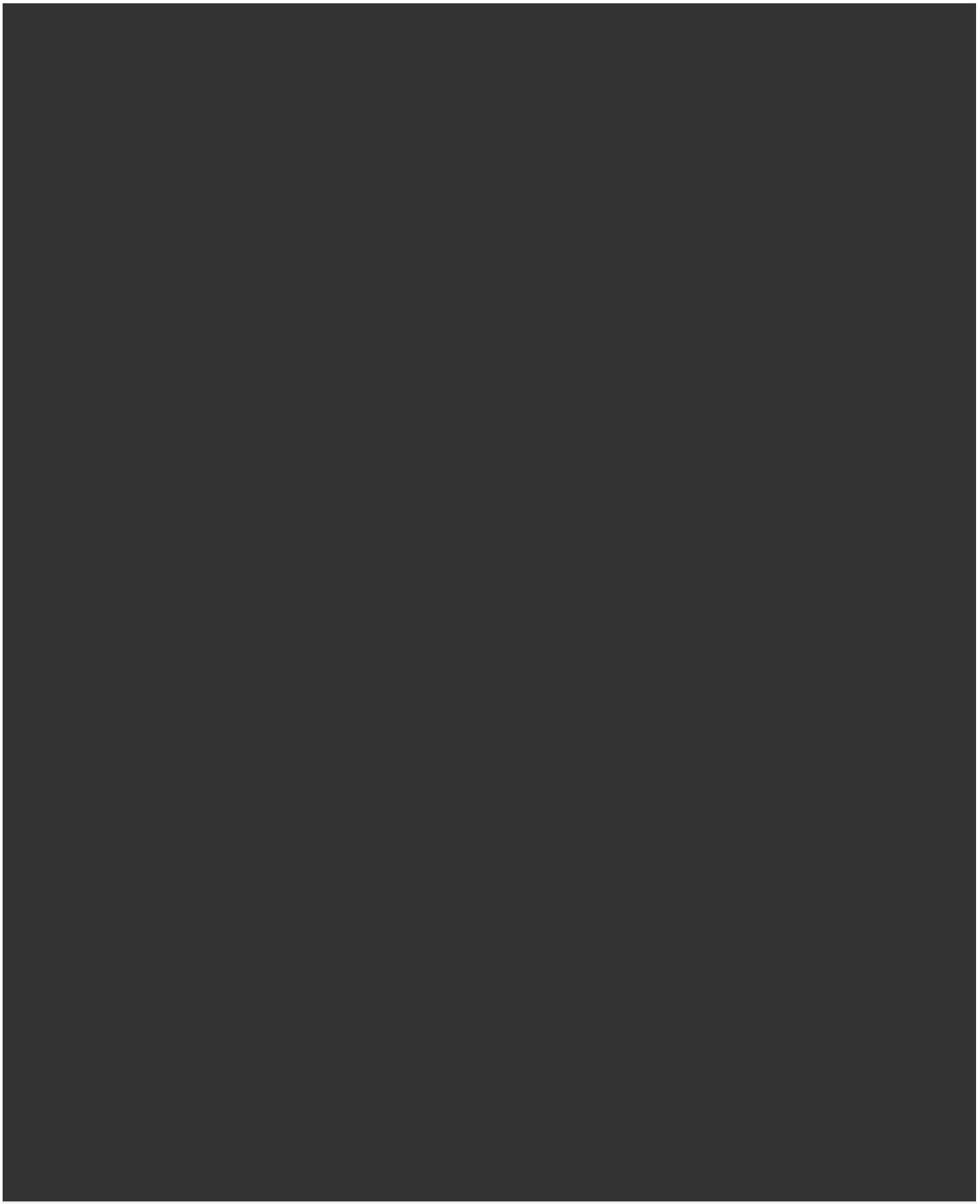
Be able to check convexity, calculate an update, and describe all minima of a simple example.

Q & A


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY ARE THERE SLAVES IN THE BIBLE
WHY ARE THERE ANTS IN MY LAPTOP
WHY ARE THERE DINOSAUR GHOSTS
WHY ARE THERE FEMALE MR MIMES
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY IS PSYCHIC WEAK TO BUG
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY ARE THERE GHOSTS
WHY ARE THERE SQUIRRELS
WHY IS SEX SO IMPORTANT
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY ARE THERE SO MANY AVENGERS
WHY ARE MY EARS RINGING
WHY DO CHILDREN GET CANCER
WHY IS THERE ICE IN SPACE
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDERS COME INSIDE
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY DO I FEEL DIZZY
WHY ARE THERE WEEKS
WHY DO I SQUINT
WHY DO SQUIRRELS DIE
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY IS THERE LAVA
WHY IS PSYCHIC WEAK TO BUG
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY IS THERE NO KING IN ENGLAND
WHY ARE DOGS AFRAID OF FIREWORKS
WHY ARE THERE SQUIRRELS
WHY IS THERE HELL IF GOD FORGIVES
WHY IS LIFE SO BORING
WHY IS GPS FREE
WHY IS LYING GOOD
WHY IS ISOLATION BAD
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS THERE NO GPS IN LAPTOPS
WHY DO SPIDER BITES ITCH
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE THERE CELEBRITIES
WHY ARE THERE TWO SUSHIES AFTER HTTP
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WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SHOSHOSTEXE RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY DO I SAY UH
WHY ARE THERE MIRRORS ABOVE BEDS
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WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND PROBES DEFENSE
WHY IS STEALING WRONG
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XKCD





Speaker notes

